Exp: 8 Nmap - Discovering Live Hosts using ARP, ICMP and TCP/UDP Scans (tryhackme)

Aim:

To discover live hosts in a network using Nmap scans (ARP, ICMP, TCP/UDP) on the TryHackMe platform.

Requirements:

* TryHackMe account and the Nmap Live Host Discovery room
* Attack box / VPN connection with Nmap installed

Procedure:

1. Log in to TryHackMe, open the room and start the AttackBox / connect through VPN.
2. Use the commands below against the target range given in the room (replace MACHINE_IP / range):

```
nmap -sn MACHINE_IP/24                  # ping scan (ARP on local LAN, ICMP/TCP otherwise)
nmap -PR -sn MACHINE_IP/24              # ARP scan
nmap -PE -sn MACHINE_IP/24              # ICMP echo scan
nmap -PP -sn MACHINE_IP/24              # ICMP timestamp scan
nmap -PM -sn MACHINE_IP/24              # ICMP address-mask scan
nmap -PS22,80,443 -sn MACHINE_IP/24     # TCP SYN ping
nmap -PA80 -sn MACHINE_IP/24            # TCP ACK ping
nmap -PU53 -sn MACHINE_IP/24            # UDP ping
nmap -sL MACHINE_IP/24                  # list scan (reverse DNS only)
```

3. Note the hosts reported as Host is up for each scan type and answer the room's questions.

Result:

Live hosts were discovered using Nmap ARP, ICMP and TCP/UDP scans on the TryHackMe platform.